August 2, 2026

SOVRA

Dear Hiring Team,

I am writing to apply for the Machine Learning Scientist position at SOVRA. I am a Master of Science in Computer Science graduate from the University of Calgary with hands-on experience in Python, PyTorch, machine learning experimentation, data-processing pipelines, backend APIs, and research communication. SOVRA's work applying AI to public procurement is appealing to me because it combines complex unstructured data, practical recommendation and information-retrieval problems, and software that supports transparent public-sector decision-making.

My graduate research focused on building reliable machine learning and data infrastructure for large, complex datasets. In my DEM super-resolution project, I trained and evaluated PyTorch ResNet models, built data-collection and preprocessing workflows with Streamlit, Rasterio, and Google Earth Engine, and assessed model quality with metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors. This work strengthened my ability to design experiments, compare model behavior, build reproducible workflows, and communicate tradeoffs clearly.

I also developed a DGGS-based framework for real-time multiresolution management of spatiotemporal Earth observation data through my BigGeo/Mitacs research collaboration. Although the domain was geospatial rather than procurement intelligence, the core problems align with SOVRA's needs: organizing large datasets, designing retrieval-oriented data structures, validating results, optimizing performance, and turning research ideas into working software. My first-author publication from this work reflects my ability to carry a technical project from design and implementation through peer-reviewed communication.

Beyond research, I have backend and NLP-adjacent engineering experience. As a Back-End Developer at Asr Gooyesh Pardaz, I developed Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL, built authentication workflows, implemented payment-related backend logic, and contributed to an AI-enabled chatbot system using GPT-style responses and FAQ data. More recently, I have been building an agentic job-search automation platform using LLM workflows, Google Sheets, Google Drive, Docker-based tooling, and workflow automation.

I would bring SOVRA a strong foundation in Python machine learning, PyTorch model development, data pipelines, retrieval-oriented systems, backend APIs, and clear technical communication. I am especially interested in contributing to systems that extract useful structure from unstructured text and help users find timely, relevant opportunities through data-driven intelligence.

Thank you for considering my application. I look forward to the opportunity to discuss how my background can support SOVRA's machine learning and market intelligence work.

Sincerely,
Amir Mirzai Golpayegani
amirgolpaa24@gmail.com